LIORESAL®

Antispastic with spinal site of action

DESCRIPTION AND COMPOSITION

Pharmaceutical form(s)

Lioresal tablets:

10 mg: White to faintly yellowish, round, flat tablets with a slightly bevelled edge. Debossed with “CG” on one side and the other the debossment “K”, score “J”. The score line on one side is to divide the tablet into equal doses.

Certain dosage strengths and dosage forms may not be available in all countries.

Active substance

Baclofen

Excipients

10 mg tablet: wheat starch; cellulose microcrystalline; povidone; silica colloidal anhydrous; magnesium stearate.

Pharmaceutical formulations may vary between countries.

INDICATIONS

Spasticity of the skeletal muscles in multiple sclerosis.

Spastic conditions occurring in spinal cord diseases of infectious, degenerative, traumatic, neoplastic, or unknown origin: e.g. spastic spinal paralysis, amyotrophic lateral sclerosis, syringomyelia, transverse myelitits, traumatic paraplegia or paraparesis, and compression of the spinal cord; muscle spasticity of cerebral origin, especially where due to infantile cerebral palsy. In spasticity due to cerebrovascular accident or degenerative or neoplastic brain disease, Lioresal is less suitable owing to the likelihood of intolerance, but it may be tried if administered cautiously.

DOSAGE REGIMEN AND ADMINISTRATION

Dosage regimen

Treatment should be initiated with small doses of Lioresal and gradually increased. It is recommended that the lowest effective dose be used. The optimum dosage must be adapted to the patient’s requirements such that clonus, flexor and extensor spasms and spasticity are reduced while avoiding adverse effects as far as possible.

To prevent excessive weakness and falling, Lioresal must be used with caution when spasticity is needed to maintain upright posture and balance in locomotion or other functions. A certain degree of muscle tone and occasional spasms may be important in helping to support circulatory functions and potentially preventing deep vein thrombosis.

Close monitoring is required at the start of treatment so that potential adverse effects such as general muscle weakness, potentially abrupt loss of muscle tone (risk of falling), fatigue or confusion can be rapidly detected and the dose adjusted.

If baclofen is to be withdrawn after prolonged administration (longer than 2-3 months), the dose should be reduced gradually over the course of approximately 3 weeks except in overdose-related emergencies, or where serious adverse effects have occurred (see section WARNINGS AND PRECAUTIONS) as abrupt withdrawal is occasionally associated with anxiety, confusional states, hallucinations, cerebral seizures, dyskinesia, tachycardia or hyperthermia. Withdrawal of baclofen should be gradual due to the risk of a transient rebound effect involving increased spasm frequency and/or severity (see section WARNINGS AND PRECAUTIONS).

Continuation of treatment with Lioresal should be reconsidered if it has not proved successful within 6-8 weeks of the patient reaching the maximum dose.

Adults

Treatment should be started with a dosage of 5 mg 3 times daily, given in 3 divided doses. Depending on the response each divided dose should be cautiously increased by 5 mg every 3 days. 30-80 mg/d is considered the standard guideline value for the daily dose, but 100 to 120 mg/d may be given in isolated cases (with monitoring in hospital).

Special dosage instructions

Children and adolescents

Experience with Lioresal is limited in children and adolescents. The minimum age for treatment is 12 months.

Treatment usually starts at very low dosages (e.g. 0.3 mg/kg/d, given in 4 divided doses). The dosage should be cautiously increased at approximately 1-2 week intervals until it meets the individual patient’s requirements. The daily dosage for maintenance therapy is 0.75-2 mg/kg. The total daily dose for children under 8 years of age must not exceed a maximum of 40 mg/day. In children over 8 years of age a maximum daily dose of 60 mg/day may be given.

Elderly patients (aged 65 years or above) and patients with spasticity of cerebral origin

The dosage should be increased particularly slowly in elderly and weakened patients suffering from organic brain disorders, cardiovascular disorders, respiratory impairment or hepatic/renal impairment and in patients with spasticity of cerebral origin.

Use in renal impairment

Lioresal should be used with caution in patients with renal impairment. As baclofen plasma concentrations are elevated in patients undergoing chronic haemodialysis, a particularly low dose of Lioresal (approx. 5 mg/d) should be selected. Symptoms of overdose have been reported with doses of over 5 mg per day in this clinical situation (see OVERDOSE).

Lioresal should not be used in patients with end stage renal failure unless the expected benefit of further treatment outweighs the potential risk. These patients must be closely monitored for early signs of overdose (e.g. drowsiness, lethargy; see sections WARNINGS AND PRECAUTIONS and OVERDOSE).

Method of administration

Lioresal tablets should be taken at mealtimes with a small amount of liquid.

CONTRAINDICATIONS

Known hypersensitivity to baclofen or to any of the excipients.

WARNINGS AND PRECAUTIONS

Psychiatric and nervous system disorders

Patients suffering from psychotic disorders, schizophrenia, depressive or manic disorders, confusional states or Parkinson's disease, should be treated cautiously with Lioresal and kept under careful surveillance, because these conditions may become exacerbated.

Suicide and suicide-related events have been reported in patients treated with baclofen. In most cases, the patients had additional risk factors associated with an increased risk of suicide including alcohol use disorder, depression and/or a history of previous suicide attempts. Close supervision of patients with additional risk factors for suicide should accompany therapy with Lioresal. Patients (and caregivers of patients) should be alerted about the need to monitor for clinical worsening, suicidal behavior or thoughts or unusual changes in behavior and to seek medical advice immediately if these symptoms present.

Epilepsy

Special attention should be given to patients known to suffer from epilepsy since lowering of the convulsion threshold may occur and seizures have occasionally been reported in connection with the discontinuation of Lioresal or with overdosage. Adequate anticonvulsive therapy should be continued and the patient should be carefully monitored.

Encephalopathy

Cases of encephalopathy have been reported in patients treated with Lioresal (see sections ADVERSE DRUG REACTIONS and OVERDOSAGE). Symptoms included somnolence, depressed level of consciousness, confusion, myoclonus and coma which are usually reversible after treatment discontinuation. If any signs or symptoms of encephalopathy are observed, baclofen should be discontinued.

Others

Lioresal should be used with caution in patients with, or with a history of, peptic ulcers, as well as in patients with cerebrovascular diseases or with respiratory or hepatic impairment.

Since adverse effects are more likely to occur, a cautious dosage schedule should be adopted in elderly and patients with spasticity of cerebral origin (see section DOSAGE REGIMEN AND ADMINISTRATION).

Renal impairment

Lioresal should be used with caution in patients with renal impairment and should be administered to end-stage renal failure patients only if the expected benefit outweighs the potential risk (see section DOSAGE REGIMEN AND ADMINISTRATION).

Neurological signs and symptoms of overdose including clinical manifestations of toxic encephalopathy (e.g., confusion, somnolence, hallucination) have been observed in patients with renal impairment taking Lioresal at doses of more than 5mg per day. Patients with renal impairment should be closely monitored for prompt diagnosis of early signs and symptoms of toxicity (see section OVERDOSAGE).

Particular caution is required when combining Lioresal with drugs or medicinal products which may significantly impact renal function. Renal function should be closely monitored and Lioresal daily dosage adjusted accordingly to prevent baclofen toxicity.

Besides discontinuing treatment, unscheduled hemodialysis might be considered as a treatment alternative in patients with severe baclofen toxicity. Hemodialysis effectively removes baclofen from the body, alleviates clinical symptoms of overdose and shortens the recovery time in these patients.

Urinary disorders

On Lioresal treatment, neurogenic disturbances affecting the emptying of the bladder may show an improvement. In patients with pre-existing sphincter hypertonia, acute retention of urine may occur; the drug should be used with caution in such cases.

Laboratory tests

In rare instances, elevated aspartate aminotransferase, blood alkaline phosphatase and blood glucose levels in the serum have been recorded. Appropriate laboratory tests should therefore be performed periodically in patients with liver disease or diabetes mellitus in order to ensure that no drug-induced changes in these underlying diseases have occurred.

Excipients

Lioresal tablets contain wheat starch. Wheat starch may contain gluten, but only in trace amounts. Taking Lioresal tablets is therefore considered safe for people with celiac disease.

Abrupt discontinuation

Anxiety and confusional state, delirium, hallucination, psychotic disorder, mania or paranoia, convulsion (status epilepticus), dyskinesia, tachycardia, hyperthermia, rhabdomyolysis and - as a rebound phenomenon - temporary aggravation of spasticity and hypertonia have been reported following the abrupt withdrawal of Lioresal, especially after long-term medication.

Drug withdrawal reactions including postnatal convulsions in neonates have been reported after intrauterine exposure to oral Lioresal. As a precautionary measure, Lioresal administration to neonates with gradual tapering can help in controlling and preventing the withdrawal reactions. (see section PREGNANCY, LACTATION, FEMALES AND MALES OF REPRODUCTIVE POTENTIAL).

Withdrawal of baclofen following long-term use (longer than 2 to 3 months) must be gradual, extending over the course of about 3 weeks, except in overdose-related emergencies or where serious adverse effects have occurred, the treatment should always be gradually discontinued by successively reducing the dosage (over a period of approximately 1 to 2 weeks).

Driving and using machines

Lioresal may be associated with adverse effects such as dizziness, sedation, somnolence and visual impairment (see section ADVERSE DRUG REACTIONS) which may negatively affect the patient’s reaction times. Patients experiencing these adverse reactions should be advised to refrain from driving or using machines.

Posture and balance

Lioresal should be used with caution when spasticity is needed to sustain an upright posture and balance in locomotion (see section DOSAGE REGIMEN AND ADMINISTRATION).

ADVERSE DRUG REACTIONS

Adverse effects occur mainly at the start of treatment (e.g. sedation, somnolence), if the dose is increased too rapidly, or if large doses are used. They are often transitory and can be attenuated or eliminated by reducing the dosage; they are rarely severe enough to require stopping the medication. In patients with a history of psychiatric illness or with cerebrovascular disorders (e.g. stroke), as well as in elderly patients, adverse reactions may be more serious.

Lowering of the convulsion threshold and convulsions may occur, particularly in epileptic patients.

Certain patients have shown increased muscle spasticity as a paradoxical reaction to the medication.

Many of the side effects reported are known to occur in association with the underlying conditions being treated.

Adverse drug reactions (Table 1) are listed according to system organ class in MedDRA. Within each system organ class, adverse reactions are ranked by frequency, with the most frequent reactions first, using the following convention (CIOMS III): very common (≥1/10); common (≥1/100, <1/10); uncommon (≥1/1,000, <1/100); rare (≥1/10,000, <1/1,000) very rare (<1/10,000), not known (cannot be estimated from the available data). Within each frequency grouping, adverse reactions are ranked in order of decreasing seriousness.

Table 1 Tabulated summary of adverse drug reactions

Immune system disorders	
Not known:	Hypersensitivity
Psychiatric disorders	
Common:	Confusional state, hallucination, depression, insomnia, euphoric mood, nightmare
Nervous system disorders	
Very common:	Sedation, somnolence
Common:	Dizziness, ataxia, tremor, headache, nystagmus
Rare:	Paraesthesia, dysarthria, dysgeusia
Not known	Encephalopathy
Eye disorders	
Common:	Visual impairment, accommodation disorder
Cardiac disorders	
Not known:	Bradycardia
Vascular disorders	
Common:	Hypotension
Gastrointestinal disorders	
Very common:	Nausea
Common:	Gastrointestinal disorder, constipation, diarrhoea, retching, vomiting, dry mouth
Rare:	Abdominal pain
Hepatobiliary disorders	

Rare:	Hepatic function abnormal
Skin and subcutaneous tissue disorders	
Common:	Rash, hyperhidrosis
Not known:	Urticaria, alopecia
Musculoskeletal and connective tissue disorders	
Common:	Muscular weakness, myalgia
Renal and urinary disorders	
Common:	Pollakiuria, enuresis, dysuria
Rare:	Urinary retention
Reproductive system and breast disorders	
Rare:	Erectile dysfunction
Not known:	Sexual dysfunction
Respiratory, thoracic and mediastinal disorders	
Common	Respiratory depression
General disorders and administration site conditions	
Common:	Fatigue
Very rare:	Hypothermia
Not known:	Drug withdrawal syndrome* (see section WARNINGS AND PRECAUTIONS), swelling face and peripheral oedema
Investigations	
Common	Cardiac output decreased
Not known:	Blood glucose increased

* Drug withdrawal syndrome including postnatal convulsions has also been reported after intra-uterine exposure to oral Lioresal

INTERACTIONS

Observed interactions to be considered

Levodopa/Dopa Decarboxylase (DDC) inhibitor (Carbidopa)

In patients with Parkinson's disease receiving treatment with Lioresal and levodopa (alone or in combination with DDC inhibitor, carbidopa), there have been reports of mental confusion, hallucinations, headaches, nausea and agitation. Worsening of the symptoms of Parkinsonism has also been reported. Hence, caution should be exercised during co administration of Lioresal and levodopa/carbidopa.

Drugs causing Central Nervous System (CNS) depression

Increased sedation may occur when Lioresal is taken concomitantly with other drugs causing CNS depression including other muscle relaxants (such as tizanidine), with synthetic opiates or with alcohol (see Driving and using machines under section WARNINGS AND PRECAUTIONS). The risk of respiratory depression is also increased. In addition, hypotension has been reported with concomitant use of morphine and intrathecal baclofen. Careful monitoring of respiratory and cardiovascular functions is essential, especially in patients with cardiopulmonary disease and respiratory muscle weakness.

Antidepressants

During concomitant treatment with tricyclic antidepressants, the effect of Lioresal may be potentiated, resulting in pronounced muscular hypotonia.

Lithium

Concomitant use of oral Lioresal and lithium resulted in aggravated hyperkinetic symptoms. Thus, caution should be exercised when Lioresal is used concomitantly with lithium.

Antihypertensives and other drugs known to lower blood pressure

Since concomitant treatment with drugs that lower blood pressure is likely to increase the fall in blood pressure, the dosage of concomitant medications should be adjusted accordingly.

Agents reducing renal function

Drugs or medicinal products that can significantly impact renal function may reduce baclofen excretion leading to toxic effects (see section WARNINGS AND PRECAUTIONS).

PREGNANCY, LACTATION, FEMALES AND MALES OF REPRODUCTIVE POTENTIALS

Pregnancy

Risk summary

There are no adequate and well-controlled studies in pregnant women. Animal data showed that baclofen crosses the placental barrier. Therefore, Lioresal should not be used during pregnancy unless the expected benefit outweighs the potential risk to the fetus.

Clinical considerations

Fetal/Neonatal adverse reactions

Drug withdrawal reactions including postnatal convulsions in neonates have been reported after intra-uterine exposure to oral Lioresal (see section WARNINGS AND PRECAUTIONS).

Animal data

Oral baclofen was shown to not have any adverse effects on fertility or postnatal development at non-maternally toxic dose levels in rats. Baclofen is not teratogenic in mice, rats, and rabbits at doses at least 2.1-times the maximum oral mg/kg dose in adults. Baclofen given orally has been shown to increase the incidence of omphaloceles (ventral hernias) in rat fetuses given approximately 8.3-times the maximum oral adult dose expressed as a mg/kg dose. This abnormality was not seen in mice or rabbits. Baclofen dosed orally has been shown to cause delayed fetal growth (ossification of bones) at doses that also caused maternal toxicity in rats and rabbits.

Lactation

In mothers taking Lioresal at therapeutic doses, the active substance passes into the breast milk, but in quantities so small that no adverse effects are to be expected in the infant.

Females and males of reproductive potential

Infertility

There is no data available on the effect of baclofen on human fertility. Baclofen did not impair male or female fertility in rats at dose levels not toxic to them.

OVERDOSAGE

Signs and symptoms

Prominent features are signs of central nervous depression or encephalopathy: somnolence, depressed level of consciousness, coma, respiratory depression.

The following symptoms may also occur: confusion hallucination, agitation, convulsion, abnormal electroencephalogram (burst suppression pattern and triphasic waves), accommodation disorder, impaired pupillary reflex, generalized muscular hypotonia, myoclonus, hyporeflexia or areflexia, peripheral vasodilation, hypotension or hypertension, bradycardia, tachycardia or cardiac arrhythmia, hypothermia, nausea, vomiting, diarrhea, salivary hypersecretion, increased hepatic enzymes, sleep apnea, rhabdomyolysis, tinnitus.

A deterioration of the overdose syndrome may occur if various substances or drugs acting on the central nervous system (e.g. alcohol, diazepam, tricyclic antidepressants) have been taken at the same time.

Treatment

No specific antidote is known.

Supportive measures and symptomatic treatment should be given for complications such as hypotension, hypertension, convulsions, gastrointestinal disturbances, and respiratory or cardiovascular depression.

Since the drug is excreted chiefly via the kidneys, generous quantities of fluid should be given, possibly together with a diuretic. Hemodialysis (sometimes unscheduled) may be useful in cases of severe poisoning associated with renal failure (see section WARNINGS AND PRECAUTIONS).

CLINICAL PHARMACOLOGY

ATC Code: M03B X01.

Mechanism of action (MOA)

Lioresal is a highly effective antispastic with a spinal site of action. Baclofen depresses monosynaptic and polysynaptic reflex transmission in the spinal cord by stimulating the GABA B receptors, this stimulation in turn inhibiting the release of the excitatory amino acids glutamate and aspartate.

Pharmacodynamics

Neuromuscular transmission is not affected by baclofen. Baclofen has an antinociceptive effect. In neurological diseases associated with skeletal muscles spasms, the clinical effects of Lioresal take the form of a beneficial action on reflex muscle contractions and marked relief from painful spasm, automatism, and clonus. Lioresal improves the patient’s mobility, making it easier to carry out daily activities (including catheterisation) and physiotherapy. Prevention and healing of decubitus ulcers, and improvement in sleep patterns (due to elimination of painful muscle spasms) and in bladder and sphincter function have also been observed as indirect effects of treatment with Lioresal, leading to a better quality of life for the patient.

Baclofen stimulates gastric acid secretion.

Pharmacokinetics

Absorption

Baclofen is rapidly and completely absorbed from the gastrointestinal tract.

Following oral administration of single doses of 10, 20, and 30 mg baclofen, peak plasma concentrations averaging about 180, 340, and 650 nanogram/mL, respectively, are recorded after 0.5 to 1.5 hours. The corresponding areas under the serum concentration curves (AUCs) are proportional to the size of the dose.

Distribution

The distribution volume of baclofen amounts to 0.7 L/kg. The protein binding is approximately 30% and is constant in the concentration range of 10 nanogram/mL to 300 microgram/mL. In the cerebrospinal fluid, the active substance attains concentrations approx. 8.5 times lower than in the plasma.

Biotransformation

Baclofen is metabolized to only a minor extent. Deamination yields the main metabolite, beta-(p-chlorophenyl)-4-hydroxybutyric acid, which is pharmacologically inactive.

Elimination/Excretion

The plasma elimination half-life of baclofen averages 3 to 4 hours. Baclofen is excreted largely in unchanged form. Within 72 hours approximately 75% of the dose is excreted via the kidneys, about 5% of this quantity being in the form of metabolites. The remainder of the dose, including 5% as metabolites, is excreted in the faeces.

Special populations

Geriatric patients (aged 65 years of age or above)

The pharmacokinetics of baclofen in elderly patients are virtually the same as in patients below 65 years of age. Following a single oral dose, elderly patients have slower elimination but a similar systemic exposure of baclofen compared to adults below 65 years of age. Extrapolation of these results to multi-dose treatment suggests no significant pharmacokinetic difference between patients below 65 years of age and elderly patients.

Pediatric patients

Following oral administration of the 2.5 mg Lioresal tablet in children (aged 2 to12 years), C_{max} of 62.8±28.7 nanogram/mL, and T_{max} in the range of 0.95 to 2 hours have been reported. Mean plasma clearance (Cl) of 315.9 mL/h/kg; volume of distribution (Vd) of 2.58 L/kg; and half-life ($T_{1/2}$) of 5.10 hours have been reported.

Hepatic impairment

No pharmacokinetic data is available in patients with hepatic impairment after administration of Lioresal. However, as the liver does not play a significant role in the disposition of baclofen, it is unlikely that baclofen pharmacokinetics would be altered to a clinically significant level in patients with hepatic impairment.

Renal impairment

No controlled clinical pharmacokinetic study is available in patients with renal impairment after administration of Lioresal. Baclofen is predominantly eliminated unchanged in the urine. Sparse plasma concentration data collected only in female patients under chronic hemodialysis or compensated renal failure indicate significantly decreased clearance and increased half-life of baclofen in these patients. Dosage adjustment of baclofen based on its systemic levels should be considered in patients with renal impairment, and prompt hemodialysis is an effective means of reversing excess baclofen in systemic circulation.

CLINICAL STUDIES

No recent clinical trials have been conducted with Lioresal.

NON-CLINICAL SAFETY DATA

Reproductive toxicity

For reproductive toxicity, see section PREGNANCY, LACTATION, FEMALES AND MALES OF REPRODUCTIVE POTENTIAL.

Mutagenicity and Carcinogenicity

Baclofen did not show any mutagenic and genotoxic potential in tests in bacteria, mammalian cells, yeast, and Chinese hamsters. The evidence suggests that baclofen is unlikely to have mutagenic potential.

Baclofen showed no carcinogenic potential in a 2-year study in rats. An apparently dose-related increase in the incidence of ovarian cysts and of enlarged and/or hemorrhagic adrenals at the maximum dose used (50 to 100 mg/kg) were observed in female rats treated with baclofen for two years.

INCOMPATIBILITIES

None known.

STORAGE

See folding box.

Lioresal should not be used after the date marked “EXP” on the pack.

Lioresal must be kept out of the sight and reach of children.

PACK SIZE

Available in a box of 10 x 5 blister packs.

INSTRUCTIONS FOR USE AND HANDLING

There are no specific instructions for use and handling.

Product Registration Holder:

Novartis Corporation (Malaysia) Sdn. Bhd.

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No. 8, Jalan SS21/37, Damansara Uptown,

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Malaysian Package Leaflet

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Novartis Pharma AG, Basel, Switzerland